

LigPlot3D: An Interactive Web-Based Framework for Geometric Analysis of Protein-Ligand Interactions

Nitin Kumar Singh*

*Discipline of Chemical Engineering, Indian Institute of Technology (IIT) Gandhinagar,
Palaj, Gujarat 382355, India*

E-mail: singh_nitin@iitgn.ac.in

Abstract

Molecular docking studies routinely generate multiple candidate binding poses, creating a practical need for rapid and interpretable inspection of protein–ligand interactions beyond numerical scoring functions. While established tools provide comprehensive interaction profiling or schematic visualization, they often require local installation or produce static outputs that limit interactive exploration. Here, this work presents *LigPlot3D*, a web-based application for interactive, client-side analysis of protein–ligand complexes. LigPlot3D combines browser-native three-dimensional molecular rendering with deterministic, geometry-based identification of key non-covalent interactions, including hydrogen bonds, salt bridges, hydrophobic contacts, π -stacking, and halogen bonds. All analysis is performed locally within the browser, ensuring data privacy and eliminating server-side dependencies. Using representative protein–ligand complexes, LigPlot3D enables rapid qualitative assessment of binding modes and interaction geometry through an intuitive interface that integrates interactive visualization

with structured interaction tables. By emphasizing accessibility, interactivity, and reproducible geometric criteria, LigPlot3D complements existing interaction profilers and visualization tools and supports efficient post-docking analysis in early-stage structure-based workflows. LigPlot3D is freely available at <https://singhnitink.github.io/LigPlot3D/>, and its source code can be cloned from GitHub and run locally using npm for offline or customized use.

Introduction

Molecular docking remains a widely used computational approach for predicting plausible binding modes and approximate affinities of small molecules in protein binding sites,¹ supporting structure-based drug discovery workflows and virtual screening pipelines. In practice, docking campaigns typically generate multiple poses per ligand (and often multiple ligands per target), creating an immediate downstream need for rapid, reliable inspection of poses, ranking consistency, and the chemical plausibility of predicted binding modes. Beyond numerical scores, the interpretability of docking results often depends on verifying whether a pose reproduces key physicochemical determinants of binding, such as hydrogen-bond patterns and local contact geometry in the binding site.

A common strategy for increasing interpretability is the explicit identification of non-covalent interactions between ligand and receptor. Automated interaction profilers such as the Protein–Ligand Interaction Profiler (PLIP) provide systematic detection and reporting of interaction types in macromolecular structures, enabling consistent characterization of binding determinants across complexes.² Recent advances have extended PLIP to incorporate protein–protein interactions, reflecting the broader demand for scalable interface analysis as structural datasets and predicted complexes continue to grow.³ These profilers provide powerful tools for analyzing molecular interactions and handling batch workflows, but they are not primarily designed as lightweight, web-first environments for quickly and interactively inspecting docking results across multiple poses and ligands.

Complementary to interaction profiling, schematic visualization tools such as LigPlot+ generate 2D interaction diagrams that summarize hydrogen bonds and non-bonded contacts, and support comparative analysis across related ligand–protein complexes.⁴ While 2D schematics can be effective for communicating interaction hypotheses and comparing related compounds, docking analysis often benefits from interactive exploration of alternative poses, quick comparison of pose-specific interactions, and easy access without requiring local installation, particularly during initial screening and iterative design.

To address these practical needs, this work presents LigPlot3D, a browser-based application for interactive analysis of docking outputs and protein–ligand complexes. All structure parsing, interaction detection, and three-dimensional rendering are performed on the client, so no structural data leaves the user’s machine and neither local installation nor server-side execution is required. LigPlot3D identifies non-covalent interactions, including hydrogen bonds, salt bridges, hydrophobic contacts, π -stacking, and halogen bonds, using explicit distance and angle criteria. Each criterion is based on a published geometric convention for the corresponding interaction type and is defined centrally within the application, making the assignments transparent and reproducible. Detected interactions are shown both in an interactive three-dimensional scene and in a structured table, allowing binding modes to be compared across different poses and ligands. The following sections describe the geometric criteria used by LigPlot3D, compare them with the default settings of an established interaction profiler, discuss how LigPlot3D relates to existing interaction-analysis tools, and evaluate the resulting assignments using a complex with independently characterized binding determinants.

Model and Methods

Architectural Framework

LigPlot3D is implemented as a fully client-side single page application, designed to prioritize interactive performance, data locality, and platform independence. All structure parsing, interaction analysis, and visualization are executed entirely within the user’s web browser, eliminating server-side dependencies and preventing external transmission of structural data. In addition to the hosted web application, users can clone the source repository from GitHub and run LigPlot3D locally using standard npm commands (`npm install` followed by `npm run dev`), enabling offline use, custom modifications, and integration into local workflows.

The application is developed using **React.js** (v19.2) with a functional component architecture. React hooks are employed for deterministic state management of molecular representations, interaction tables, and user-driven selections. The build and bundling pipeline is based on **Vite** (v7.2), which enables efficient module resolution and fast incremental rebuilds during development. Layout, typography, and responsive styling are handled using **TailwindCSS**, ensuring consistent rendering across display sizes and devices.

Molecular Visualization Engine

Three-dimensional molecular visualization is performed using a hybrid WebGL-based rendering layer that integrates **NGL Viewer** (v2.4.0)^{5,6} and **3Dmol.js** (v2.5.3).⁷ NGL Viewer is used for rendering macromolecular structures, including protein cartoons, surfaces, and backbone representations, and provides robust support for standard structural formats such as PDB and mmCIF. In parallel, 3Dmol.js is employed for efficient rendering of small molecules and localized interaction geometries.

A custom synchronization layer coordinates camera state, orientation, and selection events between the two rendering engines, allowing seamless transitions between global protein views and ligand-focused inspection while preserving spatial context.

Geometric Interaction Algorithms

LigPlot3D incorporates a deterministic, geometry-based interaction detection module intended for rapid, pose-centric analysis of protein–ligand complexes. Interaction classification is performed using explicit distance- and angle-based criteria. Each criterion is adapted from a specific, previously published geometric convention for that interaction type, cited individually in the subsections below, rather than defined ad hoc for this work. All interaction calculations are executed client-side using atomic coordinates parsed directly from the loaded structure files.

Geometric thresholds and atom classifications are defined centrally within the application configuration (`constants.ts`), ensuring transparency and reproducibility of interaction assignments. All active thresholds are documented within the application interface, allowing users to inspect the criteria underlying interaction detection.

Hydrogen Bonds

Hydrogen bonds are identified between potential donor and acceptor atoms restricted to nitrogen, oxygen, and sulfur, following the heavy-atom donor–acceptor convention established for protein structures that lack resolved hydrogens.⁸ A hydrogen bond is assigned when the donor–acceptor heavy-atom separation is less than or equal to 3.5 Å. When the donor’s hydrogen position is resolvable from the loaded structure, LigPlot3D additionally enforces a donor–hydrogen–acceptor angle criterion, following the donor–hydrogen–acceptor convention of McDonald and Thornton.⁹ LigPlot3D applies a stricter cutoff of 120° than the 90° minimum used in that work, in order to favour near-linear geometries. Because most crystallographic and docking-derived structures omit explicit hydrogens, this angle check is applied conditionally, and the heavy-atom distance criterion alone governs assignment otherwise. Fluorine atoms are explicitly excluded from the acceptor set due to their limited hydrogen-bonding capacity in biological environments.

Salt Bridges

Salt bridges are defined as electrostatic interactions between oppositely charged ionizable groups. Positively charged centers comprise the guanidinium group of arginine (NH1, NH2, CZ), the terminal nitrogen of lysine (NZ), and the imidazole nitrogens of histidine (ND1, NE2), while negatively charged centers correspond to carboxylate oxygen atoms in aspartate (OD1, OD2) and glutamate (OE1, OE2). Each charge center is taken as the centroid of its constituent atoms. A salt bridge is assigned when the distance between charge centers is less than or equal to 4.0 Å, adapted from the ion-pair distance convention reported by Barlow and Thornton.¹⁰

Hydrophobic Contacts

Hydrophobic contacts are detected between carbon atoms, including both aliphatic and aromatic carbons. All carbon atoms are considered in this analysis, regardless of their surrounding chemical groups. A distance cutoff of 4.5 Å is applied to capture van der Waals contacts that may contribute to hydrophobic stabilization within the binding site, consistent with the hydrophobic contact ranges discussed for ligand–protein recognition by Bissantz, Kuhn, and Stahl.¹¹

π -Stacking Interactions

Aromatic interactions are evaluated for residues containing planar ring systems, including phenylalanine, tyrosine, tryptophan, and histidine. Ring centroids and plane normals are computed to classify interaction geometry, following the geometric classification of aromatic side-chain pairs reported by McGaughey, Gagné, and Rappé.¹² Interactions are assigned when the centroid–centroid distance is less than or equal to 5.5 Å. Parallel stacking interactions require an inter-planar angle of 30° or less with a lateral offset not exceeding 2.0 Å, while T-shaped stacking interactions are identified by inter-planar angles of 60° or greater. LigPlot3D also detects cation– π contacts between these aromatic ring centroids and posi-

tively charged groups within 6.0 Å, the distance within which Gallivan and Dougherty found 99% of significant cation- π interactions,¹³ and reports them under the π -stacking interaction class rather than as a separate category. Cationic centres on the ligand are approximated by nitrogen atoms, without explicit protonation-state or formal-charge assignment. Across all interaction classes, the implementation relies exclusively on geometric criteria and does not incorporate energetic scoring or solvent models.

Halogen Bonds

Halogen bonds are identified between halogen donors (Cl, Br, I) and nucleophilic acceptors (N, O, S), following the directional geometric criteria for halogen bonding in biological molecules established by Auffinger and coworkers.¹⁴ An interaction is assigned when the halogen-acceptor separation is less than or equal to 3.5 Å and the C-X $\cdots$ acceptor angle, measured from the ligand carbon covalently bonded to the halogen, lies within 30° of the 165° optimum reported in that work, consistent with sigma-hole directionality. The 3.5 Å cutoff approximates their sum-of-van-der-Waals-radii criterion, which ranges from 3.27 Å for Cl $\cdots$ O to 3.50 Å for I $\cdots$ O. In contrast to the donor-hydrogen-acceptor angle used for hydrogen bonds, this criterion is measured from the carbon bearing the halogen rather than from a hydrogen, so it does not depend on explicit hydrogens being present and applies uniformly to crystallographic and docking-derived structures. In the less common case of a halogen bonded to a non-carbon atom, as in a sulfonyl chloride, no reference carbon is identified and the assignment reverts to the distance criterion alone.

Applicability Beyond Protein Receptors

Hydrogen bond, halogen bond, and hydrophobic contact detection operate on generic element types rather than residue identity, so these interaction classes already apply to ligands bound to RNA or DNA receptors to a meaningful degree. Salt-bridge and receptor-side π -stacking detection remain restricted to the canonical amino-acid residue definitions in `constants.ts`

and are not yet extended to nucleotide bases and backbone phosphate groups. Formalizing nucleic-acid-specific residue and ring definitions to bring these two interaction classes to parity is identified as a direct extension of this work.

Results and Discussion

Interactive Exploration of Binding Interfaces

LigPlot3D provides an integrated interface composed of a central three-dimensional visualization viewport and a configurable analytical sidebar. Upon loading a PDB structure, the application parses the coordinate file, identifies ligand entities, and constructs a localized representation of the binding environment. Users may selectively control protein representations, ligand styles, and interaction overlays through explicit interface controls.

The landing page allows users to drag-and-drop structure files directly into the browser window, and the overall interface layout is shown in Figure 1. Once a structure is loaded, LigPlot3D enables interactive inspection of key non-covalent interactions; a representative analysis is shown in Figure 2. Users can dynamically toggle visibility for hydrogen bonds, salt bridges, hydrophobic contacts, π -stacking (including cation- π contacts), and halogen bonds through the sidebar controls, allowing for focused analysis of specific interaction types.

Interaction Detection and Geometric Consistency

The interaction detection module was evaluated by examining representative protein-ligand complexes. For the periplasmic ligand-binding protein ProX from *Escherichia coli* in complex with proline betaine (PDB 1R9Q),¹⁵ LigPlot3D recovered the aromatic cage that coordinates the quaternary ammonium group of the ligand, detecting cation- π contacts to Trp65, Trp140, Trp188, and Tyr46, together with a salt bridge and hydrogen bonds to the ligand carboxylate (Figure 2). These assignments reproduce the binding determinants reported in the original structural analysis, reflecting the deterministic nature of the geometry-based

criteria. Notably, the two most distant of these contacts (Trp140 at 5.03 Å and Tyr46 at 5.70 Å) fall outside a 5.0 Å cutoff and are recovered only because the cation- π criterion follows the 6.0 Å convention of Gallivan and Dougherty.¹³

The application uses clear distance and angle criteria to identify interactions, helping to reduce the inclusion of weak or uncertain contacts. As a result, it produces concise interaction maps that focus on structurally meaningful interactions rather than listing every possible contact. Although LigPlot3D is not intended to capture every type of interaction, the implemented interaction classes cover the main non-covalent features that are useful for assessing docking poses.

Performance and Accessibility

All visualization and interaction analysis in LigPlot3D is executed client-side within a standard web browser. In routine use, the application loads typical protein-ligand complexes without perceptible delay and maintains smooth interactive frame rates for structures containing up to approximately 10^5 atoms. Because no server-side processing is required, LigPlot3D can be used in restricted or offline environments and does not transmit structural data externally.

This deployment model lowers the barrier to entry for docking result inspection and facilitates rapid review of binding hypotheses, making the tool suitable for early-stage pose triage, educational use, and collaborative discussion settings.

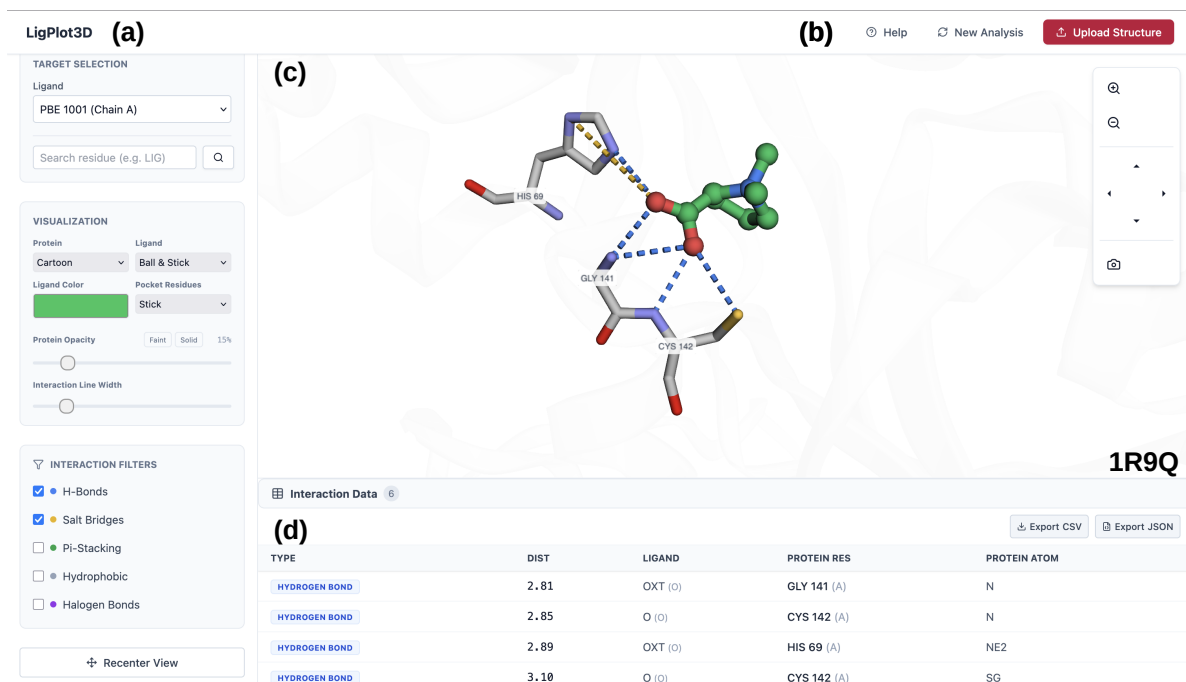


Figure 1: User interface and workflow of the LigPlot3D web application. The clean and minimal interface of LigPlot3D is organized into four functional regions to streamline protein–ligand interaction analysis. (a) Control center: The left-hand panel provides target selection and visualization controls, allowing users to select ligands, adjust representation styles (e.g., cartoon, ball-and-stick), define pocket residue display modes, customize ligand coloring, and filter interaction types such as hydrogen bonds, salt bridges, π -stacking, hydrophobic contacts, and halogen bonds. (b) Structure input module: Users can upload a local PDB file or directly enter a PDB identifier to rapidly load and visualize a protein structure, enabling seamless integration with publicly available structural data. (c) Main visualization canvas: The central 3D rendering window displays the protein–ligand complex with interactive controls for zooming, rotation, and inspection of atomic interactions. Intermolecular interactions are represented with color-coded dashed lines for intuitive interpretation. (d) Analysis pane: A structured interaction table summarizes detected contacts, including interaction type, distance metrics, ligand atom identifiers, and corresponding protein residues and atoms, facilitating quantitative evaluation and reproducible reporting of binding interactions.

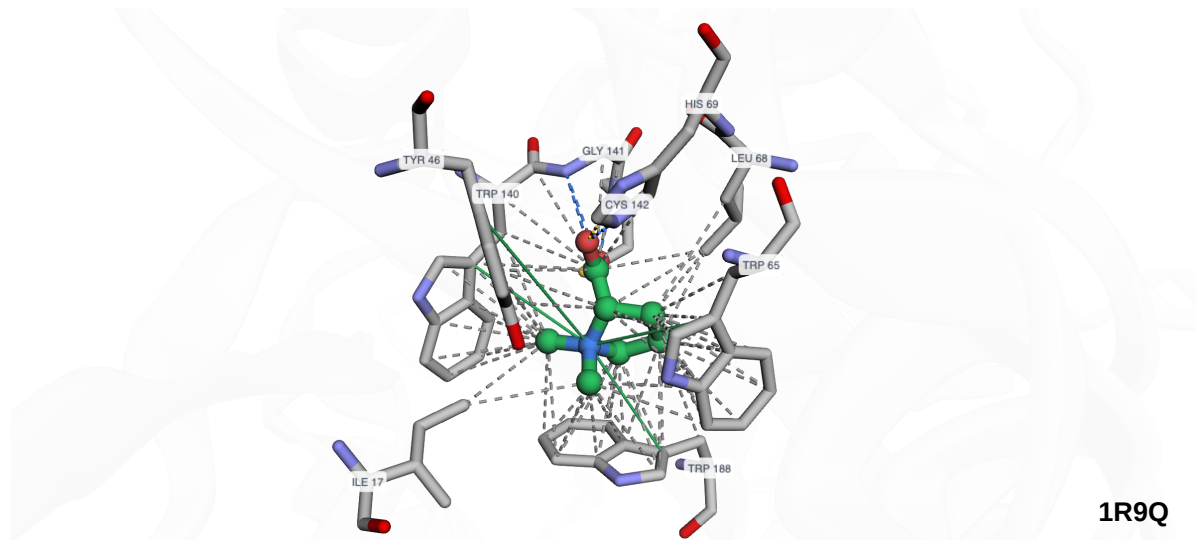


Figure 2: Interactive analysis of the ProX–proline betaine complex (PDB 1R9Q).¹⁵ LigPlot3D detects and visualizes hydrogen bonds (blue dashed lines), hydrophobic contacts (gray dashed lines), and cation– π contacts (green solidlines) to the surrounding aromatic cage in 3D; cation– π contacts are listed within the π -stacking class.

Comparison with Existing Tools

To contextualize LigPlot3D within the landscape of interaction analysis and visualization tools, its functionality was compared with LigPlot+,⁴ the Protein–Ligand Interaction Profiler (PLIP),^{2,3} Arpeggio,¹⁶ ProLIF,¹⁷ and COCOMAPS.^{18,19} Table 1 summarizes deployment model and typical use case across these tools.

LigPlot+ generates publication-quality two-dimensional interaction schematics by projecting three-dimensional complexes into flattened diagrams. These representations are effective for static reporting and comparative summaries but inherently sacrifice spatial depth information. LigPlot3D differs in that it preserves the full three-dimensional geometry of the binding site, enabling users to interrogate interaction orientations and steric relationships interactively within the browser.

PLIP provides a comprehensive and well-validated interaction profiling framework, in-

cluding interaction classes not yet implemented in LigPlot3D such as water bridges, and supports both protein–ligand and protein–protein interfaces. However, PLIP primarily operates through server-based reports or command-line workflows that generate static outputs or external visualization scripts. LigPlot3D complements this functionality by offering an interactive, client-side environment focused on rapid docking pose inspection and immediate visual feedback, without requiring data upload or local software installation. Table 2 compares LigPlot3D’s geometric thresholds against PLIP’s default configuration in detail.

Arpeggio calculates interatomic contacts directly from a structure file through an OpenBabel-based atom-typing pipeline, identifying a broad range of contact types, including ring interactions, hydrogen bonds, and steric clashes, through its web server and standalone library.¹⁶ It provides a more detailed classification of interactions than LigPlot3D, but Arpeggio is designed for single-structure and API-driven batch analysis rather than the pose-by-pose interactive comparison that LigPlot3D supports for early docking assessment.

ProLIF encodes protein–ligand and protein–protein interactions as bit-vector fingerprints using RDKit, and is built specifically for scoring and comparing large ensembles of poses or molecular dynamics frames.¹⁷ This fingerprint representation is well suited to downstream machine learning and trajectory analysis, but ProLIF is used as a Python library within a scripting or notebook environment rather than as a standalone interactive viewer, so it does not provide LigPlot3D’s direct visual inspection of a single pose.

COCOMAPS analyzes and visualizes residue-level contacts at the interface of biomolecular complexes, historically emphasizing protein–protein and protein–nucleic acid interfaces over small-molecule ligands.¹⁸ The recent COCOMAPS 2.0 release adds atomic-level interaction classification across sixteen contact types and interactive visualization,¹⁹ narrowing the visualization gap with LigPlot3D. COCOMAPS 2.0 remains a file-upload web-server workflow, however, so LigPlot3D differs by performing the equivalent client-side analysis locally, without transmitting structural data.

Table 1: Deployment model and typical use case across interaction analysis tools.

Tool	Deployment	Zero-install, no upload	Primary use case
LigPlot3D	Client-side web app	Yes	Interactive pose-by-pose docking triage
LigPlot+ ⁴	Local desktop software	No	Publication-quality 2D interaction schematics
PLIP ^{2,3}	Web server / CLI	No	Batch interaction profiling and reporting
Arpeggio ¹⁶	Web server / library	No	Detailed, API-driven contact analysis
ProLIF ¹⁷	Python library	No	Interaction fingerprints for MD and pose ensembles
COCOMAPS 2.0 ¹⁹	Web server	No	Interface contact-map analysis of biomolecular complexes

Table 2: LigPlot3D geometric thresholds compared with PLIP’s default configuration.^{2,3} The two tools are independently parameterized and their thresholds are not intended to be identical.

Interaction	LigPlot3D	PLIP (default)
Hydrogen bond distance	$\leq 3.5 \text{ \AA}$	$\leq 4.1 \text{ \AA}$
Hydrogen bond angle	$\geq 120^\circ$ (D–H–A) ^a	$\geq 100^\circ$ (donor angle)
Salt bridge distance	$\leq 4.0 \text{ \AA}$	$\leq 5.5 \text{ \AA}$
Hydrophobic contact distance	$\leq 4.5 \text{ \AA}$	$\leq 4.0 \text{ \AA}$
π -stacking distance	$\leq 5.5 \text{ \AA}$	$\leq 5.5 \text{ \AA}$
π -stacking angle deviation	$\leq 30^\circ$ (parallel)	$\leq 30^\circ$
π -stacking lateral offset	$\leq 2.0 \text{ \AA}$ (parallel)	$\leq 2.0 \text{ \AA}$
Cation– π distance	$\leq 6.0 \text{ \AA}$	$\leq 6.0 \text{ \AA}$
Halogen bond distance	$\leq 3.5 \text{ \AA}$	$\leq 4.0 \text{ \AA}$
Halogen bond angle	$165^\circ \pm 30^\circ$ (C–X $\cdots$ A)	$165^\circ \pm 30^\circ$ (donor), $120^\circ \pm 30^\circ$ (acceptor)

^aApplied when the donor’s hydrogen is resolvable from the loaded structure. Otherwise, LigPlot3D applies the heavy-atom distance criterion alone (see the Hydrogen Bonds subsection).

LigPlot3D, PLIP, Arpeggio, ProLIF, and COCOMAPS all identify interactions using geometric features rather than calculated interaction energies. Their results are therefore expected to be similar for clear contacts, such as short hydrogen bonds with nearly straight bond angles or face-to-face aromatic stacking. Differences are more likely for borderline contacts that are close to the distance or angle limits used by each tool. These differences can result from how each tool defines atom types, whether donor hydrogens are explicitly placed or inferred, and the particular distance and angle limits used by each method (Table 2). Such differences do not necessarily indicate an error in any one implementation.

However, geometric criteria alone cannot determine the strength or biological importance of an interaction. A contact detected from its geometry may be functionally important, but it could also simply result from two atoms being close together in the structure. Establishing its functional importance requires additional evidence, such as mutagenesis experiments, isothermal titration calorimetry, or quantum-mechanical calculations of interaction energies. A systematic comparison of LigPlot3D with PLIP, Arpeggio, ProLIF, and COCOMAPS using the same structures, together with comparison against experimentally validated interaction data, would be a useful next step for this work.

Conclusion

In this study, LigPlot3D is presented as a web-based application for interactive analysis of protein–ligand binding interfaces. LigPlot3D integrates browser-native three-dimensional molecular visualization with deterministic, geometry-based detection of key non-covalent interactions, enabling rapid qualitative assessment of docking poses and binding modes.

By performing all parsing, visualization, and interaction analysis client-side, LigPlot3D eliminates the need for local software installation or remote server execution, while preserving data privacy. The application is designed to complement established interaction profilers and schematic visualization tools by emphasizing interactivity, accessibility, and immediate structural feedback within a lightweight workflow.

LigPlot3D is particularly well suited for early-stage docking analysis, educational use, and rapid inspection of binding hypotheses where intuitive exploration of interaction geometry is essential. Future development will focus on extending supported interaction annotations, incorporating optional energy-related descriptors, formalizing support for RNA and DNA receptors, and enabling analysis of time-resolved structural data such as molecular dynamics trajectories.

Data and Software Availability

LigPlot3D is open-source software released under the MIT License.

- **Web Application:** <https://singhnitink.github.io/LigPlot3D/>
- **Source Code:** <https://github.com/singhnitink/LigPlot3D>

Conflicts of Interest

The author declares no conflict of interest.

References

- (1) Paggi, J. M.; Pandit, A.; Dror, R. O. The art and science of molecular docking. *Annual review of biochemistry* **2024**, *93*, 389–410.
- (2) Salentin, S.; Schreiber, S.; Haupt, V. J.; Adasme, M. F.; Schroeder, M. PLIP: fully automated protein–ligand interaction profiler. *Nucleic acids research* **2015**, *43*, W443–W447.
- (3) Schake, P.; Bolz, S. N.; Linnemann, K.; Schroeder, M. PLIP 2025: introducing protein–protein interactions to the protein–ligand interaction profiler. *Nucleic Acids Research* **2025**, *53*, W463–W465.
- (4) Laskowski, R. A.; Swindells, M. B. LigPlot+: multiple ligand–protein interaction diagrams for drug discovery. 2011.
- (5) Rose, A. S.; Hildebrand, P. W. NGL Viewer: a web application for molecular visualization. *Nucleic acids research* **2015**, *43*, W576–W579.

- (6) Rose, A. S.; Bradley, A. R.; Valasatava, Y.; Duarte, J. M.; Prlić, A.; Rose, P. W. NGL viewer: web-based molecular graphics for large complexes. *Bioinformatics* **2018**, *34*, 3755–3758.
- (7) Rego, N.; Koes, D. 3Dmol. js: molecular visualization with WebGL. *Bioinformatics* **2015**, *31*, 1322–1324.
- (8) Baker, E. N.; Hubbard, R. E. Hydrogen bonding in globular proteins. *Progress in biophysics and molecular biology* **1984**, *44*, 97–179.
- (9) McDonald, I. K.; Thornton, J. M. Satisfying hydrogen bonding potential in proteins. *Journal of molecular biology* **1994**, *238*, 777–793.
- (10) Barlow, D. J.; Thornton, J. Ion-pairs in proteins. *Journal of molecular biology* **1983**, *168*, 867–885.
- (11) Bissantz, C.; Kuhn, B.; Stahl, M. A medicinal chemist’s guide to molecular interactions. *Journal of medicinal chemistry* **2010**, *53*, 5061.
- (12) McGaughey, G. B.; Gagné, M.; Rappé, A. K. π -stacking interactions: alive and well in proteins. *Journal of Biological Chemistry* **1998**, *273*, 15458–15463.
- (13) Gallivan, J. P.; Dougherty, D. A. Cation- π interactions in structural biology. *Proceedings of the National Academy of Sciences* **1999**, *96*, 9459–9464.
- (14) Auffinger, P.; Hays, F. A.; Westhof, E.; Ho, P. S. Halogen bonds in biological molecules. *Proceedings of the National Academy of Sciences* **2004**, *101*, 16789–16794.
- (15) Schiefner, A.; Breed, J.; Bösser, L.; Kneip, S.; Gade, J.; Holtmann, G.; Diederichs, K.; Welte, W.; Bremer, E. Cation- π interactions as determinants for binding of the compatible solutes glycine betaine and proline betaine by the periplasmic ligand-binding protein ProX from *Escherichia coli*. *Journal of Biological Chemistry* **2004**, *279*, 5588–5596.

- (16) Jubb, H. C.; Higuieruelo, A. P.; Ochoa-Montaña, B.; Pitt, W. R.; Ascher, D. B.; Blundell, T. L. Arpeggio: a web server for calculating and visualising interatomic interactions in protein structures. *Journal of molecular biology* **2017**, *429*, 365–371.
- (17) Bouysset, C.; Fiorucci, S. ProLIF: a library to encode molecular interactions as fingerprints. *Journal of cheminformatics* **2021**, *13*, 72.
- (18) Vangone, A.; Spinelli, R.; Scarano, V.; Cavallo, L.; Oliva, R. COCOMAPS: a web application to analyze and visualize contacts at the interface of biomolecular complexes. *Bioinformatics* **2011**, *27*, 2915–2916.
- (19) Chawla, M.; Kalra, U.; Petta, A.; Sharma, S.; Shaikh, A. R.; Cavallo, L.; Oliva, R. COCOMAPS 2.0: a web server for identifying, analyzing, and visualizing atomic interactions at the interface of biomolecular complexes. *Bioinformatics* **2025**, *41*, btaf606.